

Historically underrepresented college students and natural mentoring relationships: A systematic review

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ARTICLE INFO

Keywords:

Natural mentoring relationships
Historically underrepresented students
College students
Review

ABSTRACT

Historically underrepresented college students experience barriers during college. One source of support that may help students manage these challenges is natural mentors. However, research on historically underrepresented college students' natural mentoring relationships is limited. The aim of this systematic review was to examine studies of historically underrepresented college students' natural mentoring relationships during college and to provide recommendations for future research. The research questions that guided the review were (a) What are the characteristics of these relationships? (b) What factors are associated with these relationships during college? and (c) Are the outcomes associated with having these relationships moderated by type of mentor? Examined studies demonstrated that natural mentors offer a variety of support in various aspects of students' lives. Relationships were linked to multiple benefits for this population, including psychological well-being, positive academic outcomes, improved self-concept, increased motivation, and development of interpersonal skills broadly. This review showed that there are still many gaps in the literature on the nuances of these relationships and the development of the relationships for historically underrepresented college students.

1. Introduction

Natural mentors—defined as caring non-parental adults who provide guidance and support—can play a positive role for adolescents and young adults throughout their development (Van Dam et al., 2018). Natural mentoring relationships tend to be long-term, close, supportive relationships and are developed organically in contrast to formal mentoring relationships (Zimmerman et al., 2005). Mentors provide college students emotional support, assist them in setting goals and choosing a career path, offer information that can help them advance their education and careers, and serve as role models (Crisp & Cruz, 2009). Historically underrepresented college students, who face a multitude of institutional barriers (Engle & Obrien, 2007), may benefit from the support of these individuals as they navigate higher education and prepare for their future careers. However, research on historically underrepresented college students' natural mentoring relationships is limited. Thus, the aims of this systematic review were to compile empirical studies on this topic and provide recommendations for future research.

1.1. Historically underrepresented students in the U.S higher education system

Historically underrepresented college students are students who are low-income, the first in their family who will receive a bachelor's degree (first-generation), and/or from an ethnic/racial group that has been historically underserved (i.e., African American/Black, Hispanic/Latinx, Pacific Islander, Native American, and East Asian) in higher education. Postsecondary educational institutions have done a poor job of retaining students from these marginalized backgrounds compared to students who have not been historically underrepresented. Specifically, first-generation college students are less likely to persist during their first two years in college (Ishitani, 2016; Lohfink & Paulsen, 2005; Radunzel, 2021) and have lower rates of earning a degree within six years after enrollment compared to continuing-generation college students (Cataldi et al., 2018). Furthermore, White students tend to have higher graduation rates compared to students of color (National Center for Education Statistics, 2019), and a similar gap exists between low-income students and their wealthier, more privileged counterparts (Yuen, 2019).

Historically underrepresented students are not a monolith; many have multiple intersecting identities which are associated with unique

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forms of marginalization in higher education (Redford & Mulvaney Hoyer, 2017). Thus individuals from these various groups tend to experience similar challenges, and research that focuses on one of these groups may have implications for students from other underrepresented groups (Nguyen and Nguyen, 2018; Redford & Mulvaney Hoyer, 2017). For instance, research on first-generation college students may be applicable to students of color due to the high proportion of Black and Brown students who are also the first in their families to receive a bachelor's degree (Nguyen and Nguyen, 2018; Redford & Mulvaney Hoyer, 2017).

Research shows that underrepresented students encounter multiple barriers in higher education. For example, students of color experience interpersonal and institutional racism, and the effects of racism (e.g., feeling isolated) have negative implications for their persistence through college (Banks & Dohy, 2019). First-generation college students experience barriers that can be attributed to the lack of knowledge on how to navigate college. In addition, low-income students have challenges covering the cost of tuition due to unmet financial aid, leading them to work long hours outside of school to cover their living expenses or may even decide to take fewer classes (Engle & O'Brien, 2007; Walizer, 2018). In sum, historically underrepresented students face various barriers, which can often be compounded due to their overlapping identities (Nguyen and Nguyen, 2018). Although natural mentors cannot entirely ameliorate oppressive institutional structures, they may be key sources of support to cope with these challenges.

1.2. Natural mentoring relationships

Rhodes' Model of Youth Mentoring (2005) posits that mentoring relationships with a strong foundation based on mutuality, trust, and empathy can promote positive youth developmental outcomes through youth's social-emotional, cognitive, and identity development. Specifically, the model suggests mentors promote social-emotional growth by helping youth develop healthy emotion regulation skills, support their cognitive development by encouraging new ways of thinking, and help youth develop their identity through role modeling (Rhodes, 2005).

Based on Rhodes' model (2005), Hagler (2018) developed a specific mentoring model for underrepresented college students. Hagler posits that natural mentoring relationships with institutional agents (i.e., adults in positions of authority who possess social and cultural capital within educational institutions) promote historically underrepresented students' educational persistence and success. Drawing upon Stanton-Salazar's (2011) social capital framework on institutional agents, Hagler theorizes that institutional agents serving as natural mentors could use their capital to help students' social-emotional, cognitive, and identity development in ways that are helpful in succeeding in college.

Specifically, Hagler suggests that institutional natural mentors could support these students' social-emotional development by offering them emotional support and providing them an opportunity to become more trusting of adults and/or institutional agents (e.g., faculty). Institutional natural mentors could also benefit students' cognitive development by offering information that is helpful in navigating educational institutions and by helping them develop an academic identity that will make them feel that they are part of the institution. Hagler (2018) suggests that through this identity development, students will be able to further develop mentoring relationships with additional institutional agents and succeed in their education. Through growth in these three developmental areas, underrepresented college students will be more likely to acquire additional support and mentoring from other institutional agents (Hagler, 2018). Although Hagler's Model focuses on how institutional natural mentors could support underrepresented college students, natural mentors outside of higher education (e.g., family members, neighborhood/community members) can also provide support during college. In fact, it is suggested that individuals have a network of mentors (Varga & Zaff, 2018) as a variety of mentors can provide a student a well-rounded system of support. For example,

mentors on campus (e.g., professor) may provide students information on soft skills (e.g., emailing a professor, time management) that are needed to be successful in college. Mentors outside of college, such as family members (e.g., grandparents, older siblings) or individuals from their community (e.g., religious leaders, neighbors), may provide encouragement to stay motivated in school.

Research has found substantial evidence in support of mentoring. For example, a meta-analysis of 30 studies of natural mentoring among youth ages 13–24 years demonstrated that the presence and quality of these relationships promote positive academic, psychological, physical health, and social-emotional development (Van Dam et al., 2018). Also notable was the finding that the *mentor role type* was significantly associated with these outcomes (Van Dam et al., 2018). Having a natural mentor in a helping profession (e.g., teacher, guidance counselor) was associated with stronger outcomes compared to other types of mentors (e.g., familial; Van Dam et al., 2018). These findings suggest the value of having mentoring relationships with institutional agents (e.g., faculty, staff) among historically underrepresented college students.

Literature reviews about mentoring relationships (both formal and informal) of undergraduate students broadly note the benefits of mentors for college students (Crisp & Cruz, 2009; Crisp et al., 2017; Jacobi, 1991). Although the authors of these reviews acknowledge the importance of mentoring underrepresented students, none of these reviews have specifically focused on the role of natural mentors in the lives of underrepresented college students. Thus, the aim of this systematic review was to synthesize empirical studies on this topic.

1.3. Current study

Historically underrepresented students face various challenges in college and may reap the benefits of having a natural mentor while in college. The goal of this systematic review is to examine the existing literature on the natural mentoring relationships of underrepresented college students. The specific questions addressed are: (a) What are the characteristics of underrepresented college students' natural mentors and mentoring relationships? (b) What factors are associated with these relationships during college? and (c) Are the outcomes associated with these relationships moderated by type of mentor?

2. Method

This systematic review adheres to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) checklist (PRISMA, 2020).

2.1. Eligibility criteria

To be included in the review, studies had to meet the following requirements: (1) include an examination of informal or naturally occurring mentoring relationships formed in the absence of any formal structure, intervention, or program (e.g., Summer Research Opportunity Program, Research Experiences for Undergraduates program, university-based mentoring programs), (2) the sample must include individuals who are in college at the time of the study, (3) a focus on historically underrepresented college students in the U.S. (i.e., students who are low-income, whose primary caregiver(s) did not graduate from a 4-year university, or who are from underrepresented racial/ethnic minority group including African America /Black, East Asian, Hispanic/Latinx, Native American/Alaskan Native, Native Hawaiian/Other Pacific Islander), or include results that focus on a subgroup of participants who are from one of the historically underrepresented groups, and (4) have been published through a peer-review process. The review excluded studies that focused on other samples of underrepresented college students, such as research that focused on students who have neurodevelopmental disorders (e.g., autism). Studies were not constrained by a specific date range.

2.2. Search strategy

Databases were chosen through consultation with a university social science reference librarian and included: (1) APA PsychInfo, (2) ERIC, (3) Professional Development Collection, and (4) Education Research Complete. Previous natural mentoring reviews (e.g., [Sánchez et al., 2018](#); [Van Dam et al., 2018](#)) were examined to develop an initial list of search terms. Afterward, the list was shared with a university social science reference librarian, and more terms were added. Search terms included undergraduate students, college student*, mentor*, natural mentor*, informal mentor*, important nonparental adult, supportive non-parental adult*, VIP*, naturally occurring mentorship*, outcomes and predictor*, and NOT “mentor* program*” or “mentor* initiatives” or “peer mentor*.” The database search was done with articles published before March 2021.

Other searches were also conducted. First, reference lists of the articles that met study criteria from the above database search were screened. Second, studies that cited the selected articles from the initial screening process were screened. Furthermore, additional hand-searching was conducted through expert consultation via a youth mentoring listserv and known researchers in the field of natural mentoring. Lastly, newly published articles were found via Google Scholar Alerts. These additional searches continued until August 2021.

2.3. Screening and selection

The data management tool Covidence (2018) was used to manage both the abstract/title screening and the full-text screening process. See

[Fig. 1](#) for the study selection process. The full screening process was conducted by the first author. The electronic database search yielded 518 peer-reviewed articles, and four additional articles were located through other sources (i.e., Google Scholar alerts, references from a mentoring researcher). During the title and abstract review, 448 articles were excluded because they did not meet the inclusion criteria (e.g., the focus of the study was on a mentoring program). During the full-text review phase, 54 articles were examined, and 45 were excluded for multiple reasons, such as the mentors reported in studies were not conducted with students in the U.S., and the focus of the studies was not on natural or informal mentoring relationships. The author consulted with four researchers who are experts on mentoring to make final decisions about the inclusion of studies. Ten articles met the study criteria.

2.4. Data extraction

A data extraction form was applied to the articles. The data extraction form was used to gather information about: (1) the aim of the study, (2) the definition used for natural mentors, (3) the study design and sample, (4) the findings and conclusions of the work.

3. Results

3.1. Study descriptions

[Table 1](#) reports a summary of included studies. The aim of the studies varied. Researchers examined factors associated with having a natural mentor in college ([Le et al., 2021](#); [Monjaras-Gaytan et al., 2021](#)), factors

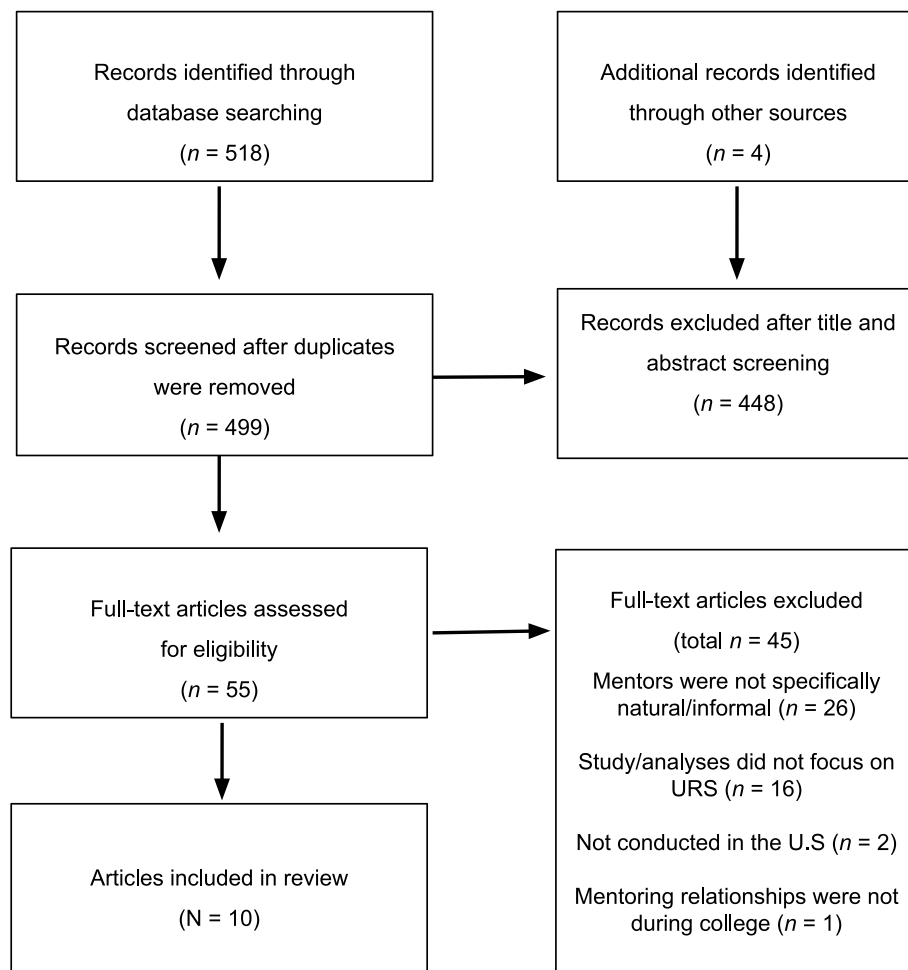


Fig. 1. PRISMA Diagram of Review Process.

Table 1

Included studies examining natural mentoring relationships of historically underrepresented college students.

Study	Study Design & Sample	Mentor and Relationship Characteristics	Key Findings
Albright & Hurd, 2018	Quantitative, longitudinal. N = 340 First year underrepresented college students at a PWI.	Not reported.	More support from a mentor (T1) was significantly correlated with more peer support (T1), parental support (T1), and anxiety symptoms (T1), but was not significantly correlated to depressive symptoms (T1/T2), anxiety symptoms (T2), or perceived stress (T1/T2).
Griffith et al., 2019	Qualitative, semi-structured interviews. N = 12 Black third- and fourth-year college students at a PWI, and their N = 10 natural mentors.	100 % of sample had natural mentors 4 were Family members 4 were People from campus 1 was a Fraternity brother's gf 1 was a White Neighbor	Students used natural mentors as a source of support to cope with race related stressors that they experienced at a PWI. Natural mentors connected students to resources and suggested different coping strategies to deal with race related stressors in college.
Hurd et al., 2016	Quantitative, longitudinal. N = 336 First year underrepresented college students at a PWI.	Fall Semester 67 % (n = 224) reported a natural mentor 33 % were High school staff 10 % were University staff 57 % were Other Spring Semester 71 % (n = 238) reported a natural mentor 21 % High school staff 18 % were University Staff 61 % Other	The total number of natural mentors was not associated with depressive symptoms, anxiety symptoms, or GPA across time. However, the number of retained natural mentors predicted reductions in depressive and anxiety symptoms from the Fall to Spring semester. And the total number of retained natural mentors was indirectly related to higher Spring GPA via reductions in depressive symptoms.
Hurd et al., 2018	Quantitative, longitudinal. N = 340 1st year underrepresented college students at a PWI.	67 % (n = 228) reported a natural mentor. 16 % had 1 mentor 17 % had 2 mentors 15 % had 3 mentors 14 % had 4 + mentors 5 % did not report number of mentors	Cumulative appraisal support was not directly associated with depressive or anxiety symptoms but was indirectly associated via improvements in perceptions of self-worth. University mentors provided more appraisal support to students compared to staff from students' former high schools; however the

Table 1 (continued)

Study	Study Design & Sample	Mentor and Relationship Characteristics	Key Findings
Le et al., 2021	Quantitative, longitudinal. N = 275 1st year college students at a PWI.	45 % were Relatives/family friend 23 % were University staff 16 % were High school staff. 16 % were Others 78 % reported a natural mentor. 46.8 % reported having 1 + natural mentor.	proportion of academic mentors did not moderate the influence of appraisal support on self worth, anxiety, or symptoms of depression. Non-underrepresented students reported more natural mentors and had more weak ties (i.e., mentoring from community members, high school staff, university staff) compared to underrepresented students.
Monjaras-Gaytan et al., 2021	Quantitative, cross-sectional. N = 521 Underrepresented college students at a PWI.	21 % (n = 110) reported an on-campus mentor. 37 % (n = 196) reported an off-campus mentor 6.7 % (n = 35) reported both an off- and on- campus mentor.	Age, help-seeking attitudes toward adults, quality of instructor relationships, having an off-campus mentor, and sense of belonging, were associated with having an institutional natural mentor. Conversely, student status (part-time vs full-time) and help-seeking avoidance behaviors was not significantly associated with having an institutional natural mentor.
Morales, 2010	Qualitative, semi-structured interviews. N = 15 Dominican/Dominican-American, first-generation college students at a PWI.	100 % of sample had natural mentors 8 were university professors, 5 were university administrators 2 were high school teachers	Mentors provided insider knowledge in academic areas (e.g., formatting papers), instrumental support for academic/professional matters (e.g., writing a resume), and encouraged and validated students' educational plans. Mentors also supported and validated students' backgrounds (e.g., being an immigrant/child of immigrants) and experiences.
Negrete et al., 2018	Quantitative, longitudinal. N = 209 First year underrepresented college students at a PWI.	100 % of sample had natural mentors 42.5 % (n = 234) mentors retained across the 1st year of college, 57.5 % (n = 316) not retained.	More relational closeness and frequent contact were associated with a higher likelihood of mentor retention. Mentor role, similarity (gender and race match) to mentor, and length

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Table 1 (continued)

Study	Study Design & Sample	Mentor and Relationship Characteristics	Key Findings
Raposa & Hurd, 2021	Quantitative, longitudinal.	N = 550 mentors were reported across all participants. 29 % had mentor, 22 % had 2 mentors, 22 % had 3 natural mentors, 27 % had 4 + natural mentors, 46.4 % (n = 255) were Relatives/Friends 34.2 % (n = 188) were High school staff 10.4 % (n = 57) were University staff; 9.1 % (n = 50) were Other	of relationship duration were not associated with mentor retention. The frequency of support and frequency conflict was not associated with mentor retention among students who had contact with their mentor in the last 30 days.
	N = 200 Second year underrepresented college students at a PWI.	100 % of sample had natural mentors N = 367 natural mentors were reported across all participants. Year 1 Fall: 51.1 % were Relatives/Friends 23.2 % were Past School Staff 8.9 % were University Staff 16.7 % were Others Year 2 Fall: 56.0 % were Relatives/Friends 20.4 % were Past School Staff 10.0 % were University Staff 13.6 % were Others	Mentors who were high school teachers/staff were less likely to provide all four types of support (i.e., emotional, informational, appraisal, and instrumental) compared to natural mentors who were extended family or family friends during the first 3 semesters of school. University faculty/staff provided more informational support than extended family or family friends during the first 3 semesters of school. Students felt closer to natural mentors who were extended family or family friends, compared to all other social roles. All types of support predicted more closeness with one's mentor.
Sánchez et al., 2006	Qualitative, semi-structured interviews.	100 % of sample had natural mentors N = 10 Mexican/Mexican-American, first-generation college students at an urban university and their N = 6 mentors.	Students reported that their mentoring relationships developed over time. Mentors provided emotional support, role modeling, guidance, informational/experiential support, and financial support. Support was provided in different areas such as interpersonal relationships, school,

Table 1 (continued)

Study	Study Design & Sample	Mentor and Relationship Characteristics	Key Findings
			careers/jobs, self-concept, values, spirituality, finances, and substance use.

Note: Underrepresented college students were defined as, students from historically underrepresented racial/ethnic background, low-income, and/or first generation. PWI = predominantly white institutions. T1 = Time 1. T2 = Time 2. GPA = grade point average.

associated with retaining natural mentors during the first year of college (Negrete et al., 2018), a description of natural mentoring relationships in college (Griffith et al., 2019; Morales, 2010; Sánchez et al., 2006; Raposa & Hurd, 2021), and positive outcomes (e.g., psychosocial outcomes, GPA) associated with having a natural mentor in college (Albright & Hurd, 2018; Hurd et al., 2016; Hurd et al., 2018).

The contexts of the studies varied. The majority of studies (n = 7) had a broad definition of natural mentoring relationships but one study specifically asked students to report natural mentors with whom students discuss topics related to their race/ethnicity (Griffith et al., 2019), and two specifically asked about mentoring relationships with university/college institutional agents (Monjaras-Gaytan et al., 2021; Morales, 2010). Most of the study samples were students from predominantly white institutions (n = 8; Albright & Hurd, 2018; Griffith et al., 2019; Hurd et al., 2016; Hurd et al., 2018; Le et al., 2021; Monjaras-Gaytan et al., 2021; Negrete et al., 2018; Raposa & Hurd, 2021). The majority of the samples were from universities in the Eastern U.S.; seven were from the southeast (Albright & Hurd, 2018; Hurd et al., 2016; Hurd et al., 2018; Le et al., 2021; Negrete et al., 2018; Raposa & Hurd, 2021) and one from the East Coast (Morales, 2010). One study took place in the Midwest (Sánchez et al., 2006). One had a sample across two institutions, one from the Midwest and the other from the Northeast (Monjaras-Gaytan et al., 2021).

3.2. Presence and number of natural mentors

Only one study compared the mentoring relationships of underrepresented students to their counterparts. Underrepresented college students (i.e., students of color, first-generation, and low-income students) were found to have fewer natural mentors compared to non-underrepresented students (Le et al., 2021). In the same study, non-underrepresented students reported more weak ties (i.e., mentoring relationships with community members, high school coaches/teachers, university faculty/staff) than underrepresented students (Le et al., 2021).

The presence and number of mentors reported by participants varied. In one study, 100 % of participants reported having at least one natural mentor in their life (Sánchez et al., 2006). Other researchers purposely selected participants with natural mentors, and so 100 % had natural mentoring relationships (Griffith et al., 2019; Morales, 2010; Negrete et al., 2018; Raposa & Hurd, 2021). Over two-thirds of the sample had at least one natural mentor in three investigations (Hurd et al., 2016; Hurd et al., 2018; Le et al., 2021). Researchers in half of the studies gave participants the option to report more than one mentor (Hurd et al., 2016; Hurd et al., 2018; Le et al., 2021; Negrete et al., 2018; Sánchez et al., 2006). In the three studies that allowed participants to report more than one mentor, students reported an average of one and two mentors (Hurd et al., 2016; Le et al., 2021; Sánchez et al., 2006). Researchers in the other two studies did not report an average number of mentors (Hurd et al., 2018; Negrete et al., 2018).

3.3. Mentor characteristics

Natural mentors had a variety of roles. The majority of natural mentors were family members, followed by school-based mentors from either their college campus or high school (Griffith et al., 2019; Hurd et al., 2018; Negrete et al., 2018; Sánchez et al., 2006; Raposa & Hurd, 2021). Researchers in two studies that specifically examined on-campus mentoring relationships found that the most common college mentors were professors (Monjaras-Gaytan et al., 2021; Morales, 2010).

Fewer studies included additional mentor demographics (i.e., age, ethnicity/race, gender; Griffith et al., 2019; Morales, 2010; Negrete et al., 2018; Sánchez et al., 2006). In two studies, there was a higher proportion of women mentors compared to men (Griffith et al., 2019; Negrete et al., 2018), and there were more male mentors than females in two other studies (Morales, 2010; Sánchez et al., 2006). Two studies that focused on specific ethnic/racial groups (i.e., Black or Mexican/Mexican-American students) indicated that mentors reflected the students' ethnicity/race (Griffith et al., 2019; Sánchez et al., 2006). However, in the Griffith et al. (2019) study, students were asked to identify natural mentors who specifically support them with race-related stressors, and mentors in the Sánchez et al. (2006) were mainly family members. On the contrary, two other studies found that the majority of mentors were White (Morales 2010; Negrete et al., 2018). In the Morales (2010) study, all the mentors were institutional agents, and as such, the majority of mentors were White. In the Negrete et al., (2018) study, 64 % of mentors were of the same race as students.

3.4. Relationship characteristics

3.4.1. Duration, frequency of contact, closeness

Researchers described the length, retention, and development of natural mentoring relationships. Investigators noted that underrepresented college students had natural mentoring relationships that lasted a mean of 1–3 years (Negrete et al., 2018). Dominican/Dominican American male college students had mentoring relationships that lasted an average of 19 months (Morales, 2010). Researchers also examined whether mentoring relationships were retained across the first year in college (Hurd et al., 2016; Negrete et al., 2018). First-year underrepresented college students reported fewer high school-based mentors in Spring compared to their Fall semester (Hurd et al., 2016). It was also reported that students developed new mentoring relationships across the first year; they formed relationships with professionals (e.g., employers) and university natural mentors over the course of the year (Hurd et al., 2016).

Other mentoring characteristics, such as relationship closeness and frequency of contact, were described. Underrepresented college students at a predominately white institution reported closer mentoring relationships with family or family friends compared to other natural mentors (e.g., former school staff; college staff, other supportive professionals; Raposa & Hurd, 2021). Relationship closeness was found to be significantly correlated with various types of mentor support, including emotional, informational, appraisal, and instrumental (Raposa & Hurd, 2021). The frequency of contact between mentors and mentees was examined in two studies. Mexican/Mexican American students were in contact with their natural mentors at least once a week (Sánchez et al., 2006). Researchers found that underrepresented college students at a predominately white institution were in contact with their natural mentors once or twice within a 30-day period (Negrete et al., 2018).

Additional mentoring relationship characteristics were described in two studies. In a qualitative study, Mexican/Mexican American college students described their relationships to be based on a level of trust and care between them and their mentors (Sánchez et al., 2006). Researchers found that underrepresented college students experienced minimal conflict in their mentoring relationship within a 30-day period (Negrete et al., 2018).

3.4.2. Social support provided

Mentors provided various types of support to underrepresented college students. Mentor support included appraisal (Morales, 2010; Raposa & Hurd, 2021; Sánchez et al., 2006), emotional support (Griffith et al., 2019; Morales, 2010; Raposa & Hurd, 2021; Sánchez et al., 2006), informational support (Griffith et al., 2019; Morales, 2010; Raposa & Hurd, 2021; Sánchez et al., 2006), instrumental support (Morales, 2010; Raposa & Hurd, 2021; Sánchez et al., 2006), and financial support (Sánchez et al., 2006). Mentors also provided support by simply being role models (Morales, 2010; Sánchez et al., 2006). Researchers found that underrepresented college students reported that the most frequent types of mentor support within the first three academic terms of college were appraisal support, emotional support, informational support, and then instrumental support in rank order (Raposa & Hurd, 2021).

More specific examples of support were reported in other investigations. A qualitative study of 15 Dominican/Dominican-American male students found that institutional agents serving as college mentors provided students insider information on how to navigate college, validated and encouraged their educational plans, applauded their resiliency, and supported and connected with them in regard to their families' recent immigration history (Morales, 2010). Qualitative interviews with 12 Black college students from a predominately white institution revealed that mentors served as a source of support when they experienced race-related stressors (e.i., being aware of negative stereotypes, unintentional racial insults, blatant discrimination) while in school (Griffith et al., 2019). Further, mentors offered mentees coping strategies to deal with these race-related stressors, such as connecting students to other resources designed to support Black students and encouraging them to "not allow race-related stressors to detract from their academic performance" (Griffith et al., 2019, p. 130). Lastly, a qualitative study of 10 Mexican/Mexican-American college students found that mentors shared stories and experiences from their own lives as a way to provide informational and experiential support (Sánchez et al., 2006).

The social support provided by mentors varied by mentor role. A study of underrepresented college students' first three college academic terms found that mentors who were students' high school teachers and staff were less likely to provide all four types of support (i.e., emotional, informational, appraisal, and instrumental) compared to mentors who were family or family friends (Raposa & Hurd, 2021). It was also found that institutional natural mentors (i.e., university faculty and staff) provided students more informational support than familial mentors across students' first three college academic terms (Raposa & Hurd, 2021). Institutional natural mentors (i.e., university faculty and staff) also provided underrepresented college students more appraisal support compared to their high school mentors (Hurd et al., 2018).

3.5. Factors associated with natural mentoring relationships

Researchers examined factors that were associated with the presence and retention of natural mentors during college. A cross-sectional study of underrepresented college students at a predominantly white institution found that being younger, having positive help-seeking attitudes, having an off-campus mentor (e.g., family mentor, high school mentor), having positive relationships with university professors, and having a positive sense of belonging on campus were significantly associated with having an institutional natural mentor, while controlling for students' years in college (Monjaras-Gaytan et al., 2021). Student status (part-time vs. full-time) and academic help-seeking avoidance were not significantly related to having a university mentor (Monjaras-Gaytan et al., 2021).

In another study, researchers found that greater relational closeness and more frequent contact with natural mentors in college were associated with a greater likelihood of retaining those mentors (i.e., keeping the same mentor) across the first year of college (Negrete et al., 2018) among underrepresented college students. However, mentor role,

gender and racial similarity with a mentor, and relationship duration were not associated with mentor retention (Negrete et al., 2018). Further, among participants who reported contact with their mentors in the last 30 days, frequency of support and conflict were not associated with mentor retention (Negrete et al., 2018).

There are positive outcomes associated with having a natural mentor in college. Mexican/Mexican American college students reported in qualitative interviews that the benefits of having a natural mentor included improved self-esteem, increased motivation, spiritual growth, learning how to navigate romantic relationships, higher grades, and reduction in substance use (Sánchez et al., 2006). Interviews of Dominican/Dominican American male first-generation college students, who were also immigrants, revealed that mentors enhanced their self-esteem and made participants feel “pride in themselves and their achievements” (Morales, 2010, p. 396).

Researchers have also examined the role of the presence of multiple mentors and changes in mentoring relationships across the first year of college in underrepresented students’ outcomes (Hurd et al., 2016). It was found that the total number of natural mentors reported in the first-year Fall semester was not directly associated with depressive symptoms, anxiety symptoms, or grade point average (GPA) in the Spring semester while controlling for Fall GPA, Fall depression and anxiety symptoms, gender, race/ethnicity, total household income, age, extraversion, and SAT/ACT scores (Hurd et al., 2016). Students’ changes in natural mentors across their first college year (i.e., changes or retention in the number of mentors) were also not associated with mental health (i.e., depression or anxiety) or GPA across the first in college. However, the total number of retained mentors from the Fall to Spring semester significantly predicted fewer depressive and anxiety symptoms, but not GPA, across students’ first year in college. Further, the total number of retained natural mentors was indirectly associated with Spring GPA through reductions in depressive symptoms while controlling for demographic characteristics and earlier grades, achievement, and mental health symptoms.

Outcomes linked to mentor support have also been explored. Researchers found that more mentor support was significantly correlated to students’ receiving more frequent peer support and parental support, and experiencing fewer symptoms of anxiety during their first semester in college (Albright & Hurd, 2018). However, mentor support during the Fall semester of students’ first college year was not significantly correlated with students’ anxiety and other psychological symptoms (i.e., depression, perceived stress) in the Spring semester (Albright & Hurd, 2018). Appraisal support in college and psychological distress (i.e., depressive and anxiety symptoms), among underrepresented college students attending a predominantly white institution, were not significantly associated with each other across time (second to the third year in college) while controlling for extraversion, gender, race/ethnicity, family income, age, depressive and anxiety symptoms from their first year in college. (Hurd et al., 2018). However, appraisal support reduced symptoms of depression and anxiety indirectly through improvements in students’ self-worth during their second year in college (Hurd et al., 2018).

3.5.1. Moderation by mentor role

Only one study examined how mentor role type (e.g., familial, nonfamilial mentors) moderates the association between underrepresented college students’ natural mentoring relationships and positive outcomes. Hurd et al. (2018) examined whether the proportion of academic mentors moderated the association between natural mentors’ appraisal support and self-esteem among underrepresented college students; no significant findings were found.

3.6. Quality of studies

Overall the methodological quality of the studies was adequate. Researchers described their methodology (e.g., data collection

procedure, description of measures) in detail. All three qualitative studies included detailed steps on how the authors increased credibility (e.g., member checking) and dependability of the findings (e.g., inter-rater reliability) and included participant quotes to highlight study themes. Only one qualitative study did not include a reflexivity statement (Morales, 2010). In the cross-sectional, quantitative study, causation cannot be implied (Monjaras-Gaytan et al., 2021). For example, it is unclear whether help-seeking attitudes led to developing natural mentoring relationships or having a natural mentor led to higher help-seeking attitudes. The other quantitative studies were longitudinal, and the authors controlled for earlier assessments of measures or for variables that have been found to be highly correlated with the measured outcomes.

4. Discussion

Previous mentoring reviews have shown that mentoring relationships (natural and formal) can greatly benefit college students (Crisp & Cruz, 2009; Crisp et al., 2017). This is the first systematic review focused on the natural mentoring relationships of historically underrepresented college students. These students encounter challenges in completing their undergraduate degree due to existing systemic barriers in higher education (e.g., discrimination, continuous high-cost of tuition) and may reap the benefits of a natural mentor during college. This study used systematic review methodology to investigate the current literature examining natural mentoring relationships of historically underrepresented college students. Through a systematic search, ten qualitative and quantitative studies were identified that met the inclusion criteria for this review. Due to the few studies that met the criteria for the systemic review, the findings should be interpreted with caution.

Overall, the study supports previous mentoring models highlighting the benefits of mentoring relationships (Hagler, 2018; Rhodes, 2005). Findings showed that natural mentors provide historically underrepresented college students a variety of social support (e.g., emotional, informational, instrumental) in both the personal and academic aspects of their lives. The support that students receive from their mentors may depend on the type of mentor. Familial mentors were found to support underrepresented college students more holistically compared to other types of mentors during their first two years of college (Raposa & Hurd, 2021). Institutional natural mentors were found to provide more informational and appraisal support compared to familial and high school mentors, which are important types of support during college (Raposa & Hurd, 2021). Unfortunately, one study found fewer underrepresented college students reported institutional natural mentors (i.e., university faculty, staff, administrators) compared to non-underrepresented college students (Le et al., 2021). These types of natural mentors are necessary in the lives of historically underrepresented college students in order to persist and graduate from college. According to Hagler’s (2018) mentoring model for underrepresented college students and Stanton-Salazar’s (2011) social capital framework on institutional agents, institutional natural mentors are well-positioned to provide underrepresented college students with resources and knowledge to navigate college successfully.

Moreover, having a natural mentor in college was associated with multiple positive outcomes. However, quantitative findings only showed positive results when mentor support and retention of mentoring relationships were examined as predictors (Hurd et al., 2016; Hurd et al., 2018). These results support previous natural mentoring reviews that found evidence that mentor support may be more predictive of positive outcomes compared to the mere presence of a mentor (de los Reyes et al., 2021; Van Dam et al., 2018). Future studies should continue to examine factors beyond the presence of a mentor, such as retention or support of a natural mentor.

One of the understudied areas in the natural mentoring relationships of historically underrepresented college students is how these relationships develop and how they are maintained. Only one study examined

antecedents of these mentoring relationships with institutional agents (Monjaras-Gaytan et al., 2021); however, it is difficult to establish causality due to its cross-sectional design. Researchers should continue to examine these relationships and consider using Hagler's Mentoring Model (2018) as a basis. Hagler theorizes that underrepresented college students could acquire institutional natural mentors through developmental processes such as gaining knowledge on how to navigate college and adaptive help-seeking skills (2018). Examining how these relationships can be maintained should also continue to be studied. Aligned with research in the youth mentoring literature (Spencer, 2007), a study found that relationship characteristics such as feelings of closeness were associated with retaining a natural mentoring relationship (Negrete et al., 2018). This study only examined the retention of natural mentors during the first year of college, the majority being pre-college mentors. Further research is needed to determine why some underrepresented students can form and retain natural mentoring relationships while in college, particularly with institutional agents.

One study found that the number of academic mentors did not significantly moderate the association between mentors' appraisal support and students' well-being (Hurd et al., 2018). Presumably, based on Hagler's (2018) and Stanton-Salazar's (2011) social capital frameworks, underrepresented college students with institutional natural mentors would be more academically successful compared to students with other kinds of mentors. Perhaps the number of academic mentors would have made a difference if the researchers had examined academic outcomes, such as retention or grades, rather than well-being.

4.1. Limitations of research

4.1.1. Existing literature

Due to the limited studies that met the criteria, the findings from this review are preliminary. Further, all quantitative studies had a broad sample of underrepresented college students, whereas three qualitative studies specifically focused on a certain ethnic/racial group that has been historically underrepresented (i.e., Dominican/Dominican-American, Mexican/Mexican-American, Black). Although many of the findings in the quantitative and qualitative studies were aligned (i.e., types of support provided, benefits of having mentors, types of mentors reported), the qualitative results had more detail about the mentoring relationships. However, findings from the qualitative studies may not be generalizable to other underrepresented college students due to their focus on a specific group of students.

The majority of the studies in this review included underrepresented college student samples from predominantly white institutions. The natural mentoring relationships of underrepresented college students attending minority-serving institutions may be different. Further, all of the samples were from the East Coast and the Midwest. Due to the ethnic/racial, socioeconomic, and cultural differences across the U.S., mentoring relationships may differ in other geographic locations. For example, a study of natural mentoring relationships of underrepresented college students in California may include more Pacific Islander students, or a study in Florida may include more Caribbean students. The findings of this review may not reflect the experiences of other underrepresented college students who were not included in this research.

4.1.2. Current review

The current review is not without limitations. First, the results of this review are limited by the study's inclusion criteria and should be interpreted within this context. The review excluded studies that had specific samples of college students, such as neurodiverse college students. However by excluding these studies, we may have missed findings that have implications for all college students including historically underrepresented college students. In addition, different fields of study may use different terms for natural mentors that were not used in the inclusion criteria. Expansion of search terms to include keywords such as role model or peer mentor may have resulted in the identification of

additional studies. Further, our study could have included specific search terms that are reflective of individuals in each underrepresented group (i.e., Latinx students) instead of just using the search term "college students." However, due to the existing number of search terms that were already included, integrating additional search terms could have reduced our search results. Future reviews should consider adding different search terms (e.g., near-peer mentors; Latinx college students) and including studies that focus on specific samples such as neurodiverse students. Second, our study was limited to peer-reviewed studies, creating the potential for publication bias. Documents such as technical reports could have included information that was relevant to the review's research questions. Future reviews should consider including other types of documents such as technical reports. Third, this review could have benefited from having multiple screeners during the title/abstract and full-text screening phases. The first author conducted the screening phase twice to compensate for this limitation. This was done to account for studies where met criteria may have been overlooked. In addition, researchers in the mentoring field provided a second opinion on whether the final articles accurately met the study criteria. Finally, only ten studies were gathered using the inclusion criteria, but various types of searches (e.g., emailing a mentoring listserv, connecting with mentoring researchers) were conducted to make sure that no relevant articles were missed. These strategies allow for confidence that most articles that met the criteria and could answer the research questions were identified.

4.2. Future directions

This review has implications for future research examining the natural mentoring relationships of historically underrepresented college students. Researchers should conduct longitudinal research to examine how mentoring relationships develop in college and the outcomes associated with having natural mentors among historically underrepresented college students. Our review demonstrated that only one study has examined factors that lead to forming natural mentoring relationships among historically underrepresented college students and that study is cross-sectional. Longitudinal work is needed to examine how students develop natural mentoring relationships while in college — specifically to examine what are the barriers and drivers of forming these relationships. Further, the review indicated that longer longitudinal studies are warranted. Long-term studies can permit the examination of whether having natural mentors during college leads to retention and graduation among underrepresented college students as well as their transition to their careers. Thus, future studies should consider following underrepresented college students through graduation and beyond. Additional study designs such as mixed methods and more qualitative studies will help triangulate data and provide a more nuanced and richer examination of these relationships.

Researchers posit that certain natural mentors could be very influential for underrepresented college students (Hagler, 2018; Stanton-Salazar, 2011); however, this review demonstrated that this hypothesis still needs to be examined. Research is needed on the differences in outcomes (e.g., GPA, psychological well-being) by mentor role type. It may be that for underrepresented college students, institutional natural mentors are more influential on academic outcomes (e.g., university sense of belonging) that are associated with college graduation compared to other types of natural mentors (e.g., high school mentors, familial mentors). On the other hand, familial mentors may be more influential on students' self-concept and psychological well-being compared to institutional natural mentors. These potential differences have not been explored and are worth investigating to better understand whether natural mentors truly make a difference.

Lastly, future studies should provide more descriptions of natural mentors and relationship characteristics. We found this information to be inconsistently reported across studies. Providing information on the duration of mentoring relationships and the demographics of the

mentors provides context to help readers understand the study findings.

4.3. Conclusion

College students who are underrepresented due to being historically underserved in higher education encounter many barriers in their academic journey and can benefit from all the support they can get while in college. This study examined the role of one type of support, natural mentors, in the lives of underrepresented college students as they navigate college by describing how these relationships look and summarizing the impact of these relationships. Studies demonstrate that mentors may provide different types of support which may impact the benefits that students reap from these connections and the retention of the relationships. Although only a few studies have examined the role of natural mentors in underrepresented college students' lives, the existing literature indicates that these supportive relationships can be essential as students navigate college. Studies reported positive benefits in students' education, psychological well-being, and self-concept. However, not all students may have a natural mentor during college, and the formation of these relationships may depend on students' characteristics and the context in which students are situated. Although studies have suggested underrepresented students benefit from natural mentoring relationships, more research is needed to examine how these findings hold in other postsecondary contexts (e.g., minority-serving institutions, community colleges) and further examine the nuances of how these relationships develop to better inform intervention efforts.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

- Albright, J. N., & Hurd, N. M. (2018). Constellations of social support among underrepresented college students: Associations with mental health. *Applied Developmental Science*, 22(4), 258–269. <https://doi.org/10.1080/10888691.2017.1287568>
- Banks, T., & Dohy, J. (2019). Mitigating Barriers to Persistence: A Review of Efforts to Improve Retention and Graduation Rates for Students of Color in Higher Education. *Higher Education Studies*, 9(1), 118–131. <https://doi.org/10.5539/hes.v9n1p118>
- Cataldi, E. F., Bennett, C. T., & Chen, X. (2018). *First-Generation Students: College Access, Persistence, and Postbachelor's Outcomes*. *Stats in Brief*. National Center for Education Statistics.
- Crisp, G., & Cruz, I. (2009). Mentoring college students: A critical review of the literature between 1990 and 2007. *Research in Higher Education*, 50(6), 525–545. <https://doi.org/10.1007/S11162-009-9130-2>
- Crisp, G., Baker, V. L., Griffin, K. A., Lunsford, L. G., & Pifer, M. J. (2017). *Mentoring Undergraduate Students: ASHE Higher Education Report*. John Wiley & Sons.
- de los Reyes, W., Sánchez, B., Polo, A., Quiroz, A., Thursby, K., O'Donnell, A., & Monjaras-Gaytan, L. Y. (2021). Mentoring Latinx Children and Adolescents: A Systematic Review. *Adolescent Research Review*, 7(1), 1–21. <https://doi.org/10.1007/s40894-021-00156-3>
- Engle, J. (2007). Postsecondary access and success for first-generation college students. *American Academic*, 3(1), 25–48.
- Engle, J., & O'Brien, C. (2007). *Demography Is Not Destiny: Increasing the Graduation Rates of Low-Income College Students at Large Public Universities*. Pell Institute for the Study of Opportunity in Higher Education. <https://files.eric.ed.gov/fulltext/ED497044.pdf>
- Griffith, A. N., Hurd, N. M., & Hussain, S. B. (2019). "I didn't come to school for this": A qualitative examination of experiences with race-related stressors and coping responses among Black students attending a predominantly White institution. *Journal of Adolescent Research*, 34(2), 115–139. <https://doi.org/10.1177/0743558417742983>
- Hagler, M. (2018). Processes of natural mentoring that promote underrepresented students' educational attainment: A theoretical model. *American Journal of Community Psychology*, 62(1–2), 150–162. <https://doi.org/10.1002/ajcp.12251>
- Hurd, N. M., Albright, J., Wittrup, A., Negrete, A., & Billingsley, J. (2018). Appraisal support from natural mentors, self-worth, and psychological distress: Examining the experiences of underrepresented students transitioning through college. *Journal of Youth and Adolescence*, 47(5), 1100–1112. <https://doi.org/10.1007/s10964-017-0798-x>
- Hurd, N. M., Tan, J. S., & Loeb, E. L. (2016). Natural mentoring relationships and the adjustment to college among underrepresented students. *American Journal of Community Psychology*, 57(3–4), 330–341. <https://doi.org/10.1002/ajcp.12059>
- Ishitani, T. T. (2016). First-generation student's persistence at four-year institutions. *College and University*, 91(3), 22–34.
- Jacobi, M. (1991). Mentoring and undergraduate academic success: A literature review. *Review of educational research*, 61(4), 505–532.
- Le, T. P., Hsu, T., & Raposa, E. B. (2021). Effects of Natural Mentoring Relationships on College Students' Mental Health: The Role of Emotion Regulation. *American Journal of Community Psychology*. <https://doi.org/10.1002/ajcp.12504>
- Lohfink, M. M., & Paulsen, M. B. (2005). Comparing the determinants of persistence for first-generation and continuing-generation students. *Journal of College Student Development*, 46(4), 409–428. <https://doi.org/10.1353/csd.2005.0040>
- Monjaras-Gaytan, L. Y., Sánchez, B., Salusky, I., & Schwartz, S. E. (2021). Historically underrepresented college students and institutional natural mentors: An ecological analysis of the development of these relationships at predominantly White institutions. *Journal of Community Psychology*, 50(2), 823–839. <https://doi.org/10.1002/jcop.22682>
- Morales, E. E. (2010). Legitimizing hope: An exploration of effective mentoring for Dominican American male college students. *Journal of College Student Retention: Research, Theory & Practice*, 11(3), 385–406. <https://doi.org/10.2190/CS.11.3.e>
- National Center for Education Statistics. (2019, February). *Indicator 23: Postsecondary Graduation Rates*. https://nces.ed.gov/programs/raceindicators/indicator_red.asp
- Negrete, A., Griffith, A. N., & Hurd, N. M. (2018). Factors contributing to the retention of natural mentors among underrepresented students across the first year of college. *Mentoring & Tutoring: Partnership in Learning*, 26(1), 70–90. <https://doi.org/10.1080/13611267.2018.1445434>
- Nguyen, T. H., & Nguyen, B. M. D. (2018). Is the "first-generation student" term useful for understanding inequality? The role of intersectionality in illuminating the implications of an accepted—yet unchallenged—term. *Review of Research in Education*, 42(1), 146–176. <https://doi.org/10.3102/0091732X18759280>
- Preferred Reporting Items for Systematic Reviews and Meta-Analyses (2020). *Prisma 2020 Checklist*. http://www.prisma-statement.org/documents/PRISMA_2020_checklist.pdf
- Radunzel, J. (2021). Examining retention and transfer in relation to incoming student data for first-generation and continuing-generation students: What differences exist? *Journal of College Student Retention: Research, Theory & Practice*, 23(2), 214–242. <https://doi.org/10.1177/1521025118813516>
- Raposa, E. B., & Hurd, N. M. (2021). Understanding networks of natural mentoring support among underrepresented college students. *Applied Developmental Science*, 25(1), 38–50. <https://doi.org/10.1080/10888691.2018.1526635>
- Redford, J., & Mulvaney Hoyer, K. (2017). *First-generation and continuing-generation college students: A comparison of high school and postsecondary experiences*. National Center for Education Statistics.
- Rhodes, J. E. (2005). A model of youth mentoring. In D. L. DuBois, & M. J. Karcher (Eds.), *Handbook of youth mentoring* (pp. 30–43). SAGE.
- Sánchez, B., Reyes, O., & Singh, J. (2006). Makin' it in college: The value of significant individuals in the lives of Mexican American adolescents. *Journal of Hispanic Higher Education*, 5(1), 48–67. <https://doi.org/10.1177/1538192705282570>
- Sánchez, B., Hurd, N. M., Neblett, E. W., & Vaclavik, D. (2018). Mentoring for Black male youth: A systematic review of the research. *Adolescent Research Review*, 3, 259–278.
- Spencer, R. (2007). "It's not what I expected" A qualitative study of youth mentoring relationship failures. *Journal of Adolescent Research*, 22(4), 331–354.
- Stanton-Salazar, R. D. (2011). A social capital framework for the study of institutional agents and their role in the empowerment of low-status students and youth. *Youth & Society*, 43(3), 1066–1109.
- Van Dam, L., Smit, D., Wildschut, B., Branje, S. J. T., Rhodes, J. E., Assink, M., & Stams, G. J. J. (2018). Does natural mentoring matter? A multilevel meta-analysis on the association between natural mentoring and youth outcomes. *American Journal of Community Psychology*, 62(1–2), 203–220. <https://doi.org/10.1002/ajcp.12248>
- Varga, S. M., & Zaff, J. F. (2018). Webs of support: An integrative framework of relationships, social networks, and social support for positive youth development. *Adolescent Research Review*, 3(1), 1–11. <https://doi.org/10.1007/S40894-017-0076-X>
- Walizer, L. (2018). *When Financial Aid Falls Short: New Data Reveal Students Face Thousands in Unmet Need*. Center for Law and Social Policy, Inc. (CLASP).
- Yuen, V. (2019, February 20). New Insights into Attainment for Low-Income Students. Center for American Progress. <https://cdn.americanprogress.org/content/uploads/2019/02/21045717/LowIncomeStudentAttainment-brief1.pdf>
- Zimmerman, M. A., Bingenheimer, J. B., & Behrendt, D. E. (2005). Natural mentoring relationships. *Handbook of Youth Mentoring*, 143–157.